

Table of contents

Electro-Magnetic interactions/coupling (Maxwell conditions)

conservativ, static charge localisation: ...

stationary, slow charge movement:

$$\begin{aligned} \nabla \cdot \epsilon_0 \mathcal{E} = \varrho & \qquad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t} \mathcal{B} = i\omega \mathcal{B} & \quad \nabla \times \frac{1}{\mu_0} \mathcal{B} = j + \epsilon_0 \frac{\partial}{\partial t} \mathcal{E} = j - i\omega \epsilon_0 \mathcal{E} \\ & & (1) \end{aligned}$$

relativistic, fast charge movement: ...